

Лабораторная работа №4

Операционные системы

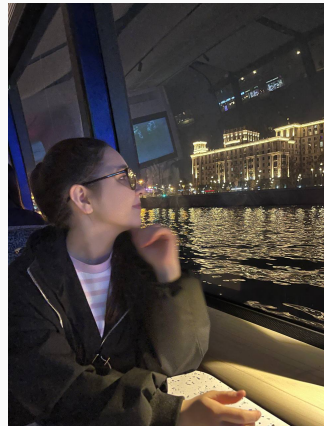
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Информация

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Цель работы

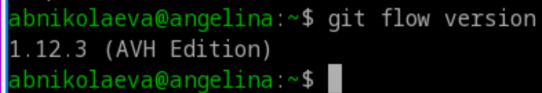
Получение навыков правильной работы с репозиториями git.

Задание

Научиться работать с git на более высоком уровне.

Выполнение лабораторной работы

Установила git-flow.

A terminal window with a black background and green text. The prompt is 'abnikolaeva@angelina:~\$'. The command 'git flow version' has been entered, and the output is '1.12.3 (AVH Edition)'. The prompt is repeated on the next line with a cursor.

```
abnikolaeva@angelina:~$ git flow version
1.12.3 (AVH Edition)
abnikolaeva@angelina:~$
```

Рис. 1: Установка git-flow

Установила Node.

```
abnikolaeva@angelina:~$ sudo dnf install nodejs
Updating and loading repositories:
Repositories loaded.
The following OpenPGP key (0x80F63AA3) is about to be removed:
Reason      : Expired on 2025-10-21 08:01:24
UserID      : "elegos_gitflow (None) <elegos#gitflow@copr.fedorahosted.org>
Fingerprint: 935799B049C5C5C36CF39A22823CA8E680F63AA3

As a result, installing packages signed with this key will fail.
It is recommended to remove the expired key to allow importing
an updated key. This might leave already installed packages unverifiable.

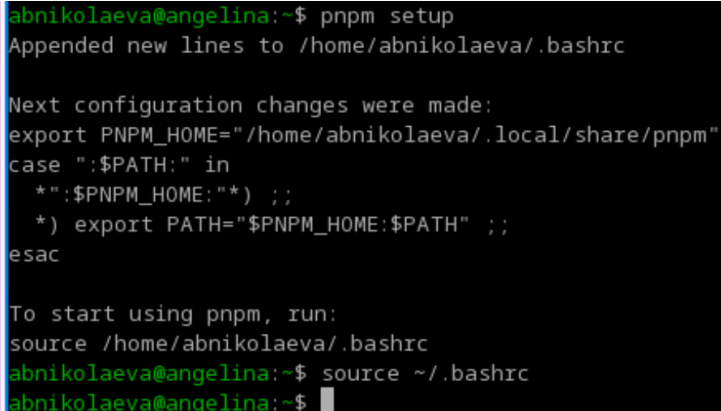
The system will now proceed with removing the key.
Is this ok [y/N]: y
Key 0x80F63AA3 was successfully removed.
```

Package	Arch	Version	Repository	Size
Installing:				
nodejs22	aarch64	1:22.22.2-3.fc44	updates	344.2 KiB
Installing dependencies:				
nodejs22-bin	noarch	1:22.22.2-3.fc44	updates	142.1 KiB
nodejs22-libs	aarch64	1:22.22.2-3.fc44	updates	73.3 MiB
nodejs22-npm-bin	noarch	1:22.22.2-3.fc44	updates	12.7 KiB
Installing weak dependencies:				
nodejs22-docs	noarch	1:22.22.2-3.fc44	updates	49.8 MiB

Рис. 2: Установка Node.js

Настройка Node.js

Запустила `pnpm setup` и выполнила `source ~/.bashrc`.

A terminal window with a dark background and light green text. The prompt is 'abnikolaeva@angelina:~\$'. The command 'pnpm setup' has been executed. The output shows that new lines were appended to the user's .bashrc file. It then displays the configuration changes made, including setting Pnpm's home directory and updating the PATH. Finally, it instructs the user to run 'source ~/.bashrc' to apply the changes, which they have done, resulting in a new prompt.

```
abnikolaeva@angelina:~$ pnpm setup
Appended new lines to /home/abnikolaeva/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/abnikolaeva/.local/share/pnpm"
case ":$PATH:" in
  *"$PNPM_HOME:") ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/abnikolaeva/.bashrc
abnikolaeva@angelina:~$ source ~/.bashrc
abnikolaeva@angelina:~$
```

Рис. 3: Запуск `pnpm setup` и выполнение `source ~/.bashrc`

Практический сценарий использования git

Подключение репозитория к github

- Создала репозиторий на GitHub. Назвала его git-extended.

The screenshot shows a GitHub profile for Angelina Nikolaeva (AngelinaNik2008). The profile includes a green and white geometric avatar, a bio, and a button to edit the profile. It also displays popular repositories: 'study_2025-2026_arch-pc', 'study_2025-2026_os-intro', and 'git-extended', all marked as public. A contribution chart shows 117 contributions in the last year, with a legend indicating the number of contributions per day (Less, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, More). The chart shows contributions in June 2026.

Popular repositories

study_2025-2026_arch-pc Public

study_2025-2026_os-intro Public

git-extended Public

117 contributions in the last year

Contribution settings

2026

Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun

Mon

Wed

Fri

Learn how we count contributions

Less More

Сделала первый коммит и выложила его на github.

```
abnikolaeva@angelina:~$ mkdir git-extendent
abnikolaeva@angelina:~$ cd git-extendent
abnikolaeva@angelina:~/git-extendent$ ls
abnikolaeva@angelina:~/git-extendent$ git init
Initialized empty Git repository in /home/abnikolaeva/git-extendent/.git/
abnikolaeva@angelina:~/git-extendent$ git commit -m "first commit"
On branch master

Initial commit

nothing to commit (create/copy files and use "git add" to track)
abnikolaeva@angelina:~/git-extendent$
```

Рис. 5: Создание первого коммита и его загрузка на github

Конфигурация общепринятых коммитов.

- Конфигурация пакетов Node.js .

```
abnikolaeva@angelina:~/git-extendent$ pnpm init
Wrote to /home/abnikolaeva/git-extendent/package.json

{
  "name": "git-extendent",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "packageManager": "pnpm@10.33.0"
```

Конфигурация git-flow

- Инициализировала git-flow.

```
abnikolaeva@angelina:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
  - master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/abnikolaeva/git-extended/.git/hooks]
abnikolaeva@angelina:~/git-extended$
```

Снимок экрана

- Проверила, что я на ветке develop, загрузила весь репозиторий в хранилище, установила внешнюю ветку как вышестоящую для этой ветки и создала релиз с версией 1.0.0.

```
abnikolaeva@angelina:~/git-extended$ git branch
* develop
  master
abnikolaeva@angelina:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/AngelinaNik2008/git-extended/pull/new/develop
remote:
To github.com:AngelinaNik2008/git-extended.git
 * [new branch]      develop -> develop
abnikolaeva@angelina:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
abnikolaeva@angelina:~/git-extended$
```

Скрин экрана

Рис. 9: Проверка нахождения на ветке develop, загрузка всего репозитория в хранилище, установка внешней ветки как вышестоящей для этой ветки и создание релиза с версией 1.0.0

Создание релиза git-flow

- Создала релиз с версией 1.2.3, обновила номер версии в `package.json`, создала журнал изменений и добавила журнал изменений в индекс, залила релизную ветку в основную ветку.

```
abnikolaeva@angelina:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

abnikolaeva@angelina:~/git-extended$ standard-changelog --first-release
abnikolaeva@angelina:~/git-extended$ git add CHANGELOG.md
abnikolaeva@angelina:~/git-extended$ git commit -am 'chore(site: add changelog)'
[release/1.0.0 9f190ee] chore(site: add changelog)
1 file changed, 7 insertions(+)
```

- Отправила данные на github, создала релиз на github с комментариями из журнала изменений.

All branches ▾

All activity ▾

All users ▾

All time ▾

Showing most recent first ▾

chore(site: add changelog) ...

 AngelinaNik2008 created [release/1.0.0](#) • 9f190ee • 2 minutes ago

Merge branch 'release/1.0.0' ...

 AngelinaNik2008 pushed 2 commits to [master](#) • 405af53...cd90a5e
• 2 minutes ago

feat: 0 ...

 AngelinaNik2008 created [develop](#) • 405af53 • 11 minutes ago

feat: 0 ...

 AngelinaNik2008 pushed 1 commit to [master](#) • 58af6e9...405af53
• 13 minutes ago

feat: 0 ...

 AngelinaNik2008 pushed 1 commit to [master](#) • 9b91cf6...58af6e9
• 15 minutes ago

first commit ...

 AngelinaNik2008 created [master](#) • 9b91cf6 • 57 minutes ago

Выводы

В ходе данной лабораторной работы были получены навыки правильной работы с репозиториями git.